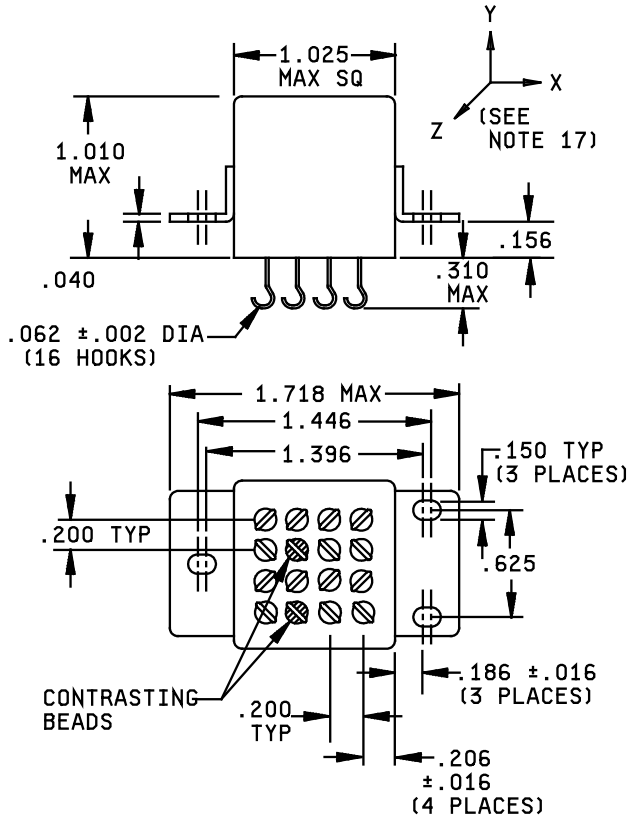
INCH-POUND
MIL-PRF-83536/19E
23 September 2021
SUPERSEDING
MIL-PRF-83536/19D
26 March 2020

PERFORMANCE SPECIFICATION SHEET

# RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, 4PDT, LOW LEVEL TO 10 AMPERES, MAGNETIC LATCH, HERMETICALLY SEALED, ALL WELDED, TRANSIENT SUPPRESSED DC COILS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-83536.

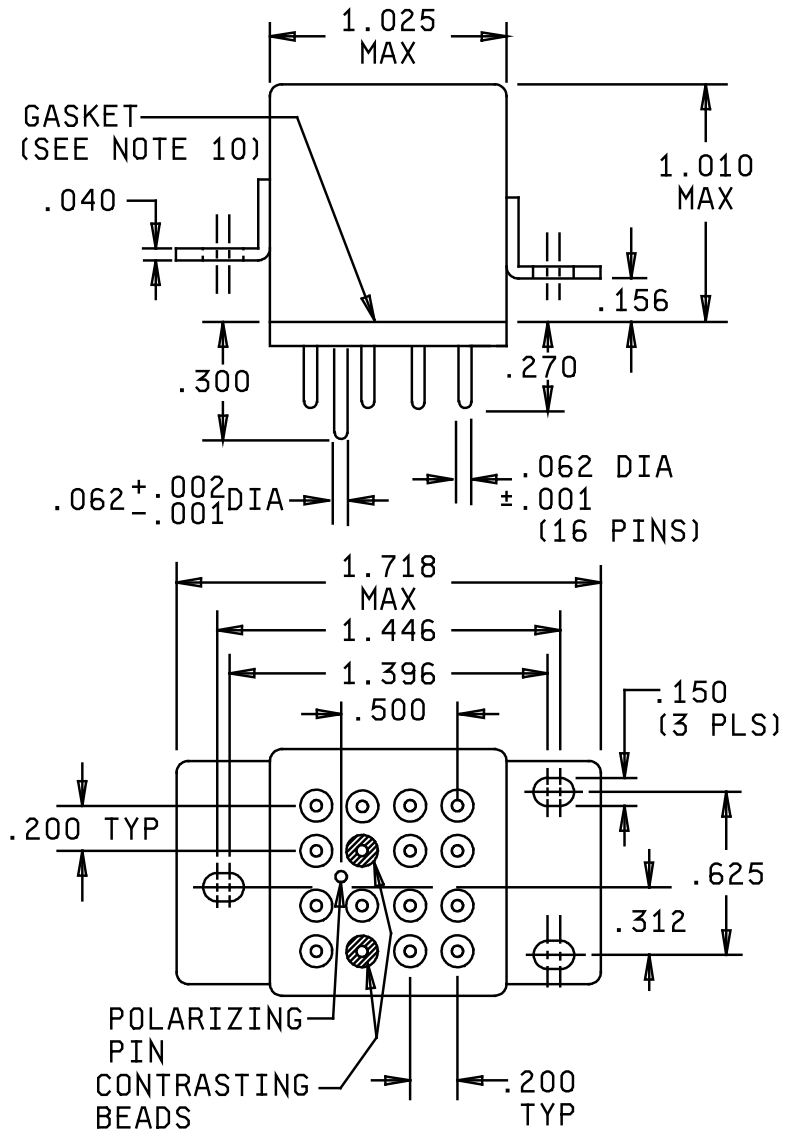



RAISED VERTICAL FLANGE MOUNT (RVFM) WITH SOLDER HOOKS

FIGURE 1. Dimensions and configurations.

AMSC N/A
FSC 5945
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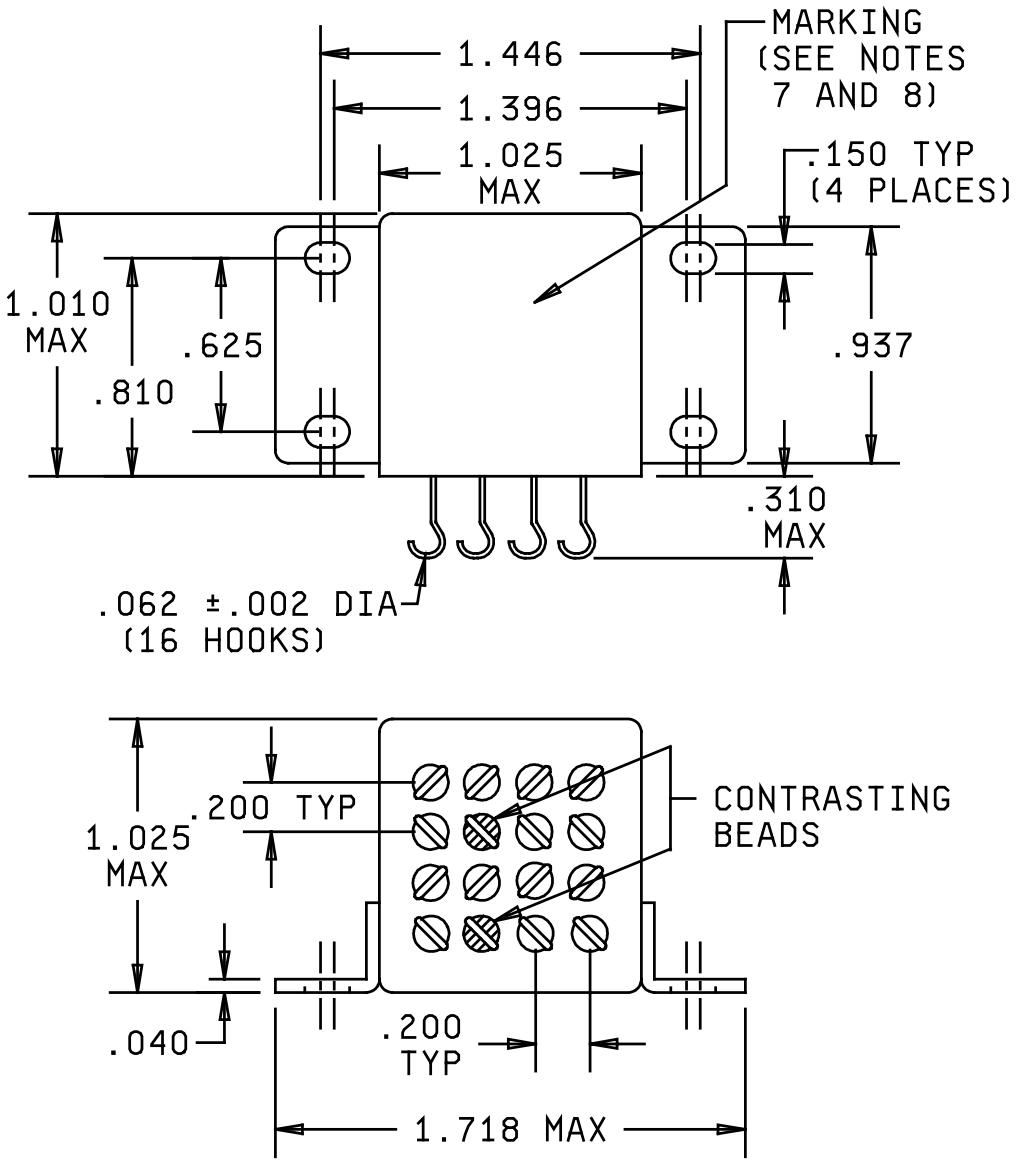



RAISED VERTICAL FLANGE MOUNT (RVFM) WITH SOCKET PINS

FIGURE 1. Dimensions and configurations - Continued.



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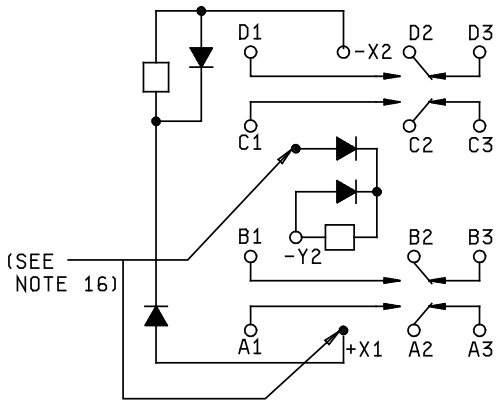



HORIZONTAL FLANGE MOUNT WITH SOLDER HOOKS

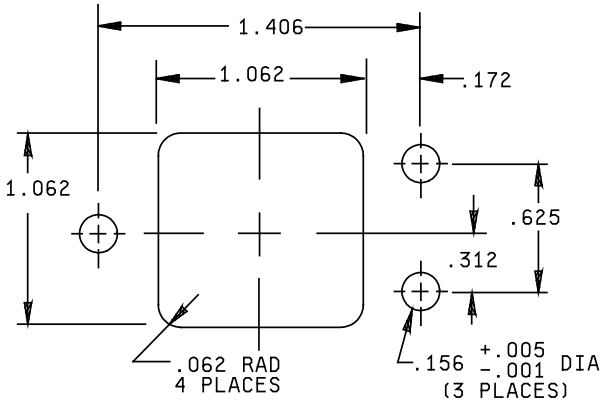
FIGURE 1. Dimensions and configurations - Continued.



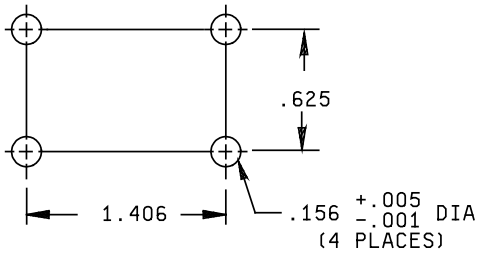
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CIRCUIT DIAGRAM




MOUNTING HOLE LAYOUT (RVFM)




MOUNTING HOLE LAYOUT (HFM)

FIGURE 1. Dimensions and configurations - Continued.



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| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .000 | 0.00 | .172 | 4.37 | .937 | 23.80 |
| .001 | 0.03 | .200 | 5.08 | 1.010 | 25.65 |
| .002 | 0.05 | .270 | 6.86 | 1.025 | 26.04 |
| .005 | 0.13 | .300 | 7.62 | 1.062 | 26.97 |
| .040 | 1.02 | .310 | 7.87 | 1.396 | 35.46 |
| .050 | 1.27 | .312 | 7.92 | 1.406 | 35.71 |
| .062 | 1.57 | .500 | 12.70 | 1.446 | 36.73 |
| .150 | 3.81 | .625 | 15.88 | 1.718 | 43.64 |
| .156 | 3.96 | .810 | 20.57 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .010 (0.25mm).
4. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
5. These relays are polarized monostable.
6. This relay shall not be damaged by reverse polarity.
7. Applicable to horizontal flange mount only. The circuit diagram, manufacturer's PIN, and the military PIN shall be marked on the near side. The remaining portion of the nameplate data shall be marked on the far side.
8. Applicable to horizontal flange mount only. Relays shall be marked with the manufacturer's name or CAGE code and date code. Marking shall be with the bottom of the print adjacent to the near side.
9. Socket pin terminals shall provide the operational, environmental, and interface characteristics to provide a reliable interconnect to gold-plated contacts. Terminals, except the polarizing pin, shall be gold plated. One system for gold plating that may be used is ASTM-B488, type 3, class 1.25 with a nickel underplate of 50 to 150 microinches thick. The gold plating system shall enable the product to meet the performance requirements of this specification and shall be approved by the qualifying activity.
10. Gasket shall provide a reliable seal between the relay and mating socket that will meet the environmental, operational, and interface requirements of the relay with the mating socket. The gasket shall have shore hardness 15 to 35, thickness .050 ±. .010. Gasket material according to SAE-AMS3332 has been considered acceptable.
11. Relay is magnetically latched in both positions.
12. All hooks shall be tin finished.
13. Relays shall have a (+) sign placed on the circuit diagram as shown.
14. Coil symbol optional in accordance with MIL-STD-1285.
15. Transient voltage (back EMF) 5 V dc maximum. Diodes shall have peak inverse voltage rating of 600 V dc.
16. Indicated terminal shall be identified by a contrasting bead.
17. Plane of critical motion for vibration and shock is Y-axis.

FIGURE 1. Dimensions and configurations - Continued.



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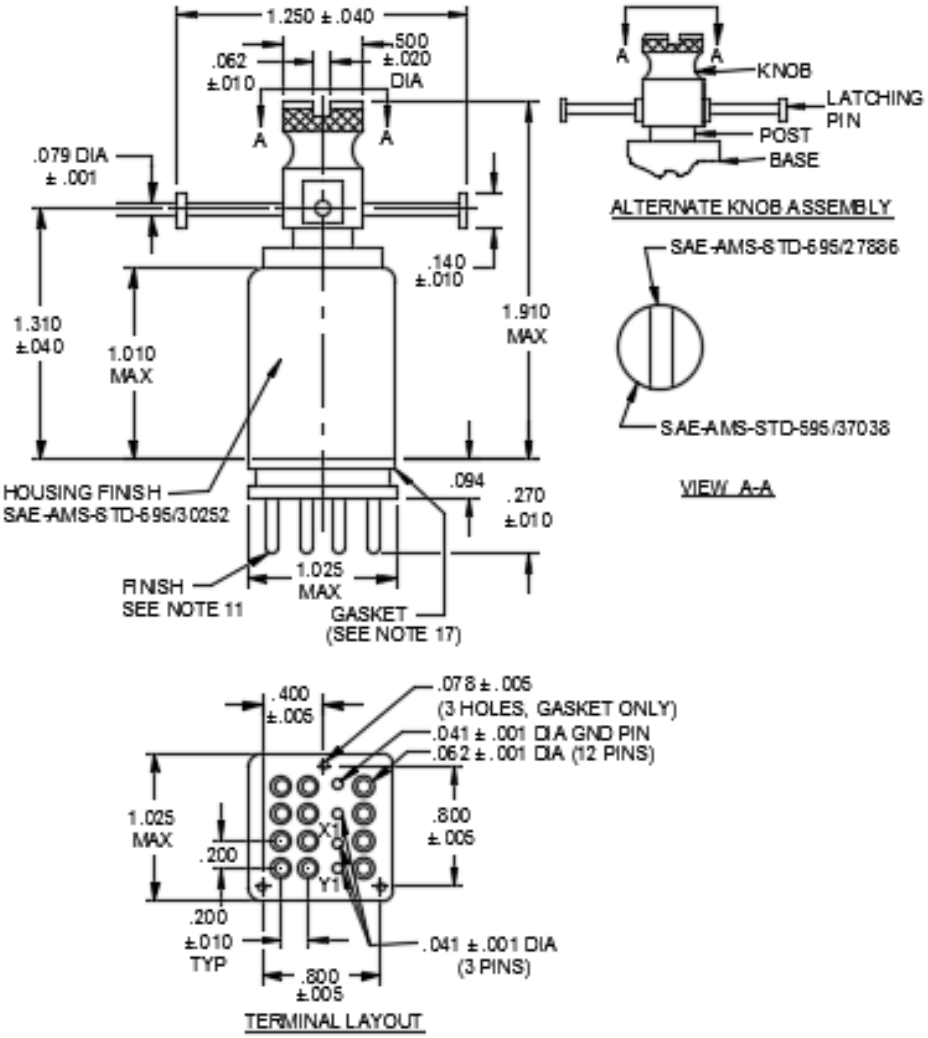
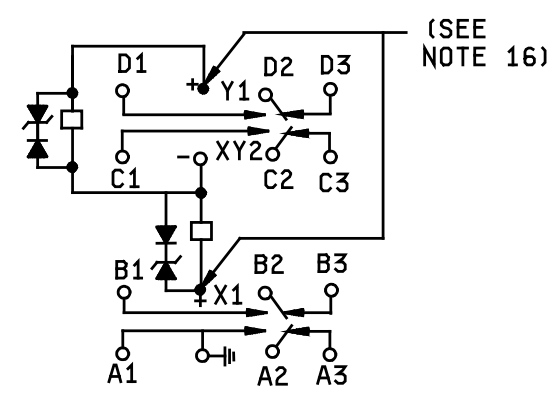



FIGURE 2. Outline dimensions and configuration, track mount (see notes 16 and 17).

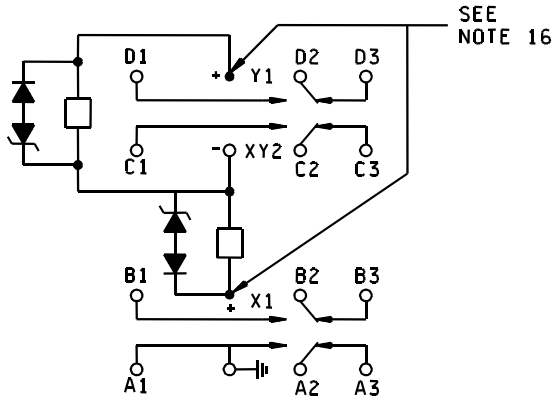


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(SEE NOTES 4, 5, AND 6)

CIRCUIT DIAGRAM (TRACK MOUNT)




SEE NOTES 4, 5, AND 6

CIRCUIT DIAGRAM (TRACK MOUNT)
(ALTERNATE INTERNAL
DIODE CONFIGURATION)

| Inches | mm | Inches | mm |
|---|---|---|---|
| .001 | 0.03 | .200 | 5.08 |
| .005 | 0.13 | .270 | 6.86 |
| .010 | 0.25 | .400 | 10.16 |
| .020 | 0.51 | .500 | 12.70 |
| .040 | 1.02 | .800 | 20.32 |
| .062 | 1.57 | 1.010 | 25.65 |
| .078 | 1.98 | 1.025 | 26.04 |
| .079 | 2.01 | 1.250 | 31.75 |
| .094 | 2.39 | 1.310 | 33.27 |
| .140 | 3.56 | 1.910 | 48.51 |

FIGURE 2. Outline dimensions and configuration, track mount (see notes 13 and 14) – Continued.



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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .010 (0.25mm) for three place decimals and ±.03 (0.76 mm) for two place decimals.
4. Relay is magnetically latched in both positions.
5. Caution note to observe polarity must appear on relays with dc coils.
6. Terminal numbers shall not appear on the header or on the gasket. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
7. Enclosure shall be brown in accordance with SAE-AMS-STD-595, color 30252.
8. With the same magnitude and polarity voltage applied to terminals X1-XY2 simultaneously, the relay contacts shall not transfer from the condition existing prior to the application of voltage to both coils.
9. With 20 dc maximum applied initially to terminals X1-XY2 the subsequent application of 28 V dc minimum to terminals Y1-XY2 (while the voltage is maintained on terminals X1-XY2) shall not cause the relay contacts to transfer from the condition resulting from the application of voltage to terminals X1-XY2 prior to the application of voltage to terminals Y1-XY2. Removal of voltage from terminals X1-XY2 (while voltage is maintained on terminals Y1-XY2) shall cause relay contact transfer.
10. With 20 V dc maximum applied initially to terminals Y1-XY2 the subsequent application of 28 V dc to terminals X1-XY2 (while the voltage is maintained on terminals Y1-XY2) shall not cause the relay contacts to transfer from the condition resulting from the application of voltage to terminals Y1-XY2 prior to the application of voltage to terminals X1-XY2. Removal of voltage from terminals Y1-XY2 (while voltage is maintained on terminals X1-XY2) shall cause relay contact.
11. Socket pin terminals shall provide the operational, environmental, and interface characteristics to provide a reliable interconnect to gold-plated contacts. Terminals, except the polarizing pin, shall be gold plated. One system for gold plating that may be used is ASTM-B488, type 3, class 1.25 with a nickel underplate of 50 to 150 microinches thick. The gold plating system shall enable the product to meet the performance requirements of this specification and shall be approved by the qualifying activity.
12. Back EMF (transient voltage): 42 V dc maximum.
13. Track mount base, knob, latching pin and post shall be corrosion resistant. One way of ensuring corrosion resistance is to nickel plate stainless steel 303 with 100 microinches minimum thickness per McDonnell Douglas P.S. 13112 (Deutsch Relays, Inc. Patent number 3.790,915) or equivalent 100 microinches thick minimum.
14. Track mount relay shall be capable of mating with M6106/5 “Bracket and Socket Assembly”, configuration III. The latching pin shall be capable of withstanding an axial pull of 35 pounds, in a direction away from the knob, without degradation. The rotation of the knob mechanism shall be restricted such that the latching pins cannot be placed on the side opposite the M6106/5 bracket cutouts.
15. JANTX or equivalent screened semiconductors shall be used for internal coil suppression. Relays using suppression devices shall continue to operate should the suppression circuit be in failure mode. Diodes shall have a peak inverse voltage of 600 V dc minimum.
16. Indicated terminals shall be identified by contrasting beads.
17. Gasket shall provide a reliable seal between the relay and mating socket that will meet the environmental, operational, and interface requirements of the relay with the mating socket. The gasket shall have shore hardness 30 to 45. Gasket material according to SAE- AMS3332 has been considered acceptable.

FIGURE 2. Outline dimensions and configuration, track mount (see notes 13 and 14) – Continued.



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REQUIREMENTS:

Contact data:

Load ratings: See table I.

Low level: 10 to 50 μA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Mixed loads: Applicable.

TABLE I. Rated contact load and cycles (amperes per pole).

| Type of load (high level) | Cycles x $10^3$ | 28 V dc | 115 V ac 1 phase 400 Hz | 115 V ac 1 phase 50/60 Hz 1/ | 115/200 V ac 3 phase 400 Hz | 115/200 V ac 3 phase 50/60 Hz 1/ |
|---|---|---|---|---|---|---|
| Resistive | 100 | 10 | 10 | 2.5 | 10 | 2.5 |
| Inductive | 20 | 8 | 8 | N/A | 8 | N/A |
| Inductive | 10 | N/A | N/A | 2.5 | N/A | 2.5 |
| Motor | 100 | 4 | 4 | 2 | 4 | 2 |
| Lamp | 100 | 2 | 2 | 1 | N/A | N/A |

1/ For 50/60 Hz rating, rupture and overload are not applicable and life for each load shall be 10,000 cycles.

Life: 100,000 cycles, unless otherwise specified (see table I).

Contact voltage drop and resistance:

Initial contact voltage drop: 0.100 volt maximum. Rated resistive current at 6 V dc or peak ac. Relays shall not make or break this load.

Initial contact resistance: 0.010 ohm maximum. 50 mA max at 6 V dc or peak ac.

High level life (contact voltage drop):

During life: 10 percent open circuit voltage maximum.

After life: 0.125 volt maximum.

Intermediate current (contact resistance):

During intermediate current: 3 ohms maximum.

After intermediate current: 0.150 ohm maximum.

Low level (contact resistance):

During low level: 100 ohms maximum.

After low level: 0.150 ohm maximum at 100 mA and 28 V dc.



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Contact bounce: 1.0 millisecond maximum.

Contact stabilization time: 2.5 milliseconds maximum.

Overload current: 40 amperes dc (60 amperes ac).

Rupture current: 50 amperes dc, (80 amperes ac).

Time current relay characteristics: See table II.

TABLE II. Time current relay characteristics. 1/

| Successive application | Amperes | Time |
|---|---|---|
| 1 | 15 | 1 hour |
| 2 | 50 | 5 seconds |
| 3 | 100 | 1.2 seconds |
| 4 | 250 | 0.2 second |
| 5 | 350 | 0.1 second |

1/ All relays shall withstand overload and fault currents. Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of five different current levels in the sequence listed in table II. Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz 3-phase. Cooling time between successive applications shall be 30 minutes. Tests shall be performed on both normally open and normally closed contacts. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion.

Coil data 1/ 2/: See table III.

Operate time: 15 milliseconds maximum (each coil), 20 milliseconds maximum (each coil) (track mount) with rated coil voltage over the temperature range.

Duty rating: Continuous.

Coil transient suppression: Applicable (42 V dc maximum for track mount and 5 V dc maximum for all others).

Neutral screen: Applicable.

1/ CAUTION: Due to possible interaction of relay magnetic fields, the following spacing requirements, as a minimum, shall be considered in dense packaging situations:

a. Row to row assisting fields: .125 (3.18 mm).

b. Row to row opposing fields: .1875 (4.763 mm).

c. Side to side alternating fields: .0625 (1.588 mm).

d. Side to side like fields: .125 (3.18 mm).

2/ Track mount relays shall have the pickup, voltage and timing tests, performed with a M6106/5 bracket placed around the relay housing.



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TABLE III. Dash numbers and characteristics. 1/

| Dash number 2/ | | Mount | Coil data | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | At 25°C | | | | Over temperature range | Maximum pickup voltage | |
| Solder hook | Socket pin | | Coil voltage (V dc) 3/ | | Coil current (amperes) maximum | Specified latch/ reset voltage (V dc ) 4/ | Specified latch/ reset voltage (V dc ) 4/ | High temperature test | Continuous current test |
| | | | Rated | Max | | | | | |
| 001 | 002 | RVFM | 6 | 7.3 | 0.300 | 3.5 | 4.5 | 5.0 | 5.7 |
| 003 | --- | HFM | 6 | 7.3 | 0.300 | 3.5 | 4.5 | 5.0 | 5.7 |
| 004 | 005 | RVFM | 12 | 14.5 | 0.150 | 6.5 | 9.0 | 9.9 | 11.25 |
| 006 | --- | HFM | 12 | 14.5 | 0.150 | 6.5 | 9.0 | 9.9 | 11.25 |
| 007 | 008 | RVFM | 28 | 29 5/ | 0.072 | 14.5 | 18.0 | 19.8 | 22.5 |
| 009 | --- | HFM | 28 | 29 5/ | 0.072 | 14.5 | 18.0 | 19.8 | 22.5 |
| --- | 010 | Track | 28 | 29 5/ | 0.072 | 14.5 | 18.0 | 19.8 | 22.5 |
| 011 | 012 | RVFM | 48 | 50 | 0.038 | 24.0 | 36.0 | 38.0 | 42.0 |
| 013 | --- | HFM | 48 | 50 | 0.038 | 24.0 | 36.0 | 38.0 | 42.0 |

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications.
2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example: 001L, 002R.
3/ CAUTION: The use of any coil voltages less than the rated coil voltage will compromise the operation of the relay.
4/ Pickup voltage as shown is for test purposes only and is not to be used for design criteria.
5/ When maximum ambient temperature does not exceed +85°C, the maximum coil voltage shall be 32 V dc.

Electrical data:

Insulation resistance:

Initial: 100 megohms minimum at 500 V dc.

After life or environmental tests: 100 megohms minimum at 500 V dc (track mount). 50 megohms minimum at 500 V dc (all others).

Dielectric withstanding voltage 3/:

| | Track mount | | All others | |
|---|---|---|---|---|
| | Coil to coil Coil to case | All other points | Coil to coil Coil to case | All other points |
| Sea level: | | | | |
| Initial: | 1050 | 1500 | 1000 | 1250 |
| After life: | 1050 | 1250 | 1000 | 1000 |
| Altitude: | | | | |
| At 80,000 feet: | 350 | 350 | 250 | 350 |
| At 300,000 feet: | N/A | N/A | 500 | 500 |

3/ Dielectric may be improved by suitable insulation of terminals and wiring after installation.



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Maximum leakage current for track mount only: 100 microamperes rms.

Environmental characteristics:

Temperature range: -70°C to +125°C.

Maximum altitude rating: 300,000 feet, 80,000 feet for track mount.

Shock (specified pulse): Applicable, MIL-STD-202-213, test condition C, 200 g's for 6 ±1 ms, except horizontal flange mount peak g value shall be 100 g's for 6 ±1 rms. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (sinusoidal): Applicable, MIL-STD-202-204, 30 g's, except frequency range shall be 10 to 3,000 Hz. Horizontal flange mount shall be 20 g's, 57 to 3,000 Hz. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (random): Applicable to track mount only. Applicable to qualification only. Test in accordance with MIL-STD-202-214, test condition IG, 1.5 hours per axis. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (random): Applicable to qualification and group C inspection. Test in accordance with MIL-STD-202-214, test condition IG (0.4 $g^2$/Hz, 50 to 2000 Hz) for horizontal flange mount, test condition IE (0.2 $g^2$/Hz) applies. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (random): Applicable to track mount only (qualification only). Test in accordance with MIL-STD-202-214, test condition IG, 1.5 hours per axis. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Acceleration: Applicable, except 15 g's.

Physical data:

Dimensions and configurations: See figure 1.

Weight: 0.17 pound (77.1 grams), 0.2 pound (90.7 grams) track mount.

Terminal strength:

Solder hook terminals:

Pull force: 10 pounds ±1.0 pound.

Bend: Applicable.

Twist: Not applicable.

Socket pin terminals:

Pull force: 10 pounds ±1.0 pound.

Bend: Applicable.

Twist: Not applicable.



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Terminal solderability: Applicable to solder hook terminals only.

Seal: Hermetic.

Marking: Applicable.

Part or Identifying Number (PIN): M83536/19 (dash number from table III and suffix letter designating failure rate level). PIN is a new term encompassing previous terms used in specifications such as part number, type designator, and identification number.

Qualification inspection:

Qualification inspection and sample size: See table IV.

TABLE IV. Qualification inspection and sample size. 1/

| Single submission | Group submission | |
|---|---|---|
| 18 units plus 1 open unit for level L at C = 0 2/<br>33 units plus 1 open unit for level M at C = 0 2/<br>Qualification inspection as applicable | M83536/19-008<br>M83536/19-008<br>M83536/19-008<br>M83536/19-007<br>M83536/19-009<br>M83536/19-028 | 18 units plus 1 open unit for level L at C = 0 2/<br>33 units plus 1 open unit for level M at C = 0 2/<br>Qualification inspection as applicable<br>2 units, qualification inspection table, group II, shock, vibration, acceleration, terminal strength, and seal |
| | M83536/19-002<br>M83536/19-006<br>M83536/19-012 | 2 units, qualification inspection table, group II |

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-PRF-83536/18 may be used in addition to MIL-PRF-83536/19 data. Prior to performance of retention of qualification testing, the relay manufacturer shall preselect the sampling plan.

2/ The number of units required for qualification testing shall be increased as required in group V, table II of MIL-PRF-83536, if the contractor elects to test the number of units permitting one or more failures. Prior to performance of qualification testing, the relay manufacturer shall preselect the sampling plan.

Supersession data: See table V.

TABLE V. Supersession data.

| Superseded PIN | Replacement PIN M83536/19- |
|---|---|
| M6106/51- 004 | 007 |
| 005 | 008 |
| 006 | 009 |
| M6106/37- 001 | 010 |



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Referenced documents. In addition to MIL-PRF-83536, this document references the following:

| | | |
|---|---|---|
| ASTM-B488 | SAE-AMS3332 | MIL-STD-1285 |
| SAE-AMS-STD-595 | MIL-STD-202-213 | MIL-STD-202-204 |
| MIL-STD-202-214 | MIL-PRF-83536/18 | |

Custodian activities:
Army-CR
Navy-EC
Air Force -85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2021-031)

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